

Neural means and kernel corrections for an EMIT atmospheric radiative-transfer emulator

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Abstract

We compare neural networks, exact Matérn regression and their combinations on a six-dimensional radiative-transfer table over the 285-band EMIT grid. Across ten random splits, with validation-based selection and 18,884 training samples, input-scaled kernels achieve 0.095% mean relative radiance error, compared with 0.76% for cubic regression and 0.39% for a neural network. Residual correction reduces the network error to 0.14%. A kernel on wider-network features reaches 0.079%, and its convex stack reaches 0.078%. This small forward-error gain accompanies an increase in the mean within-split 95th-percentile reflectance error from 7.57 to 22.71 percentage points, with a larger tail at every split. We derive exact residual-error comparisons and a conditional stability bound relating radiance to reflectance errors. Learning-curve, output-rank and width experiments characterize the effects of representation and tuning. Comparisons with OCO-2 and ten further configurations demonstrate the task dependence of the relative performance of neural and kernel emulators.

1 Introduction

Atmospheric correction for imaging spectroscopy requires repeated evaluations of a radiative-transfer model. Learned emulators reduce this computational cost by approximating the map from atmospheric and viewing conditions to spectral coefficients. Neural emulators have been incorporated into reflectance-retrieval systems [2, 1], while Gaussian-process methods provide accurate surrogates for the OCO-2 forward model [4]. Physics-guided multi-fidelity models offer another approach [3]. These applications raise a practical question: when is it preferable to fit a kernel directly on the atmospheric state, learn a neural representation, or combine both?

Kernel methods are competitive with neural architectures on a range of operator-learning benchmarks [6]. Networks such as DeepONet and Fourier neural operators learn representations for high-dimensional inputs and outputs [7, 8]; kernel regression instead specifies a metric and solves a regularized interpolation problem [11]. Their combination can take several forms. Deep kernel learning jointly trains a representation and a kernel [22], and kernel flows can regularize intermediate network layers [9]. Here we use the construction of [5]: a trained network supplies either a mean whose residual is fitted by a kernel, or fixed features on which a kernel regresses the target. We compare these predictors with direct kernel regression, polynomial regression, coordinatewise selection and convex stacking.

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The main dataset contains 23,313 radiative-transfer evaluations over the 285-band EMIT wavelength grid [10]. Its six-dimensional input and low-rank spectral output make exact kernel regression feasible at the full training size. Ten random partitions provide matched comparisons of the model families, followed by experiments varying training size, output rank, network width and kernel tuning. Input-specific length scales substantially improve the direct kernel. Wider-network features further reduce radiance error, even though the corresponding network has a larger standalone error under the tested optimizer and training budget.

The forward and inverse objectives give different model rankings. A convex stack slightly reduces the wide feature kernel’s mean radiance error, but increases its 95th-percentile reflectance error at every split. We derive exact finite-perturbation identities that relate this distinction to the denominator of the reflectance inverse. We also give an exact error-alignment criterion for residual correction and separate its kernel-regression error from principal-component truncation. These calculations complement the empirical comparison by identifying the quantities required for improvement in a specified metric.

Section 2 describes the data, evaluation protocol and inversion stability. Sections 3–5 present the models, results and ablations. Section 6 analyses residual correction. Sections 7 and 8 compare the findings with OCO-2 and ten further emulation configurations. The study concerns interpolation of a fixed radiative-transfer table; the implications for retrieval and computational cost are discussed in Section 9.

2 Data and protocol

The data are $n = 23,313$ tabulated radiative-transfer evaluations. Each input $x \in \mathbb{R}^6$ collects the atmospheric and viewing state; the coordinate ranges are consistent with the standard lookup-table axes for imaging-spectroscopy atmospheric correction [1] — an aerosol optical depth in $[0.05, 1.2]$, a surface elevation in $[-0.5, 6]$ km, a water-vapor column in $[0.05, 7]$ cm, a relative azimuth in $[0, 180]^\circ$, a solar zenith angle in $[0, 65]^\circ$, and a view zenith angle within 30° of nadir. Each output is a set of four spectral components on the same 285-point EMIT wavelength grid spanning 381–2493 nm: path radiance Y_1 , direct flux Y_2 , diffuse flux Y_3 , and spherical albedo Y_4 (Figure 1). The quantities retrieved downstream are the radiance at a test surface reflectance ρ ,

$$L(\rho) = Y_1 + \rho \frac{Y_2 + Y_3}{1 - \rho Y_4}, \quad (1)$$

and the reflectance recovered by inverting (1) with predicted components at the true radiance,

$$\hat{\rho} = \frac{L - \hat{Y}_1}{\hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)}, \quad (2)$$

both evaluated at $\rho = 0.7$.

The protocol is the same for every family. The table is split into a test block of 2,331 rows and a remaining block of 20,982; 2,098 of the latter are held out for validation, leaving 18,884 for fitting. The procedure is repeated with seeds 101–110. Standardization and principal components are fitted only on each split’s training rows. Hyperparameters, checkpoints, coordinatewise selections and combination weights use that split’s validation rows. Each fitted family is then evaluated on that split’s test rows. Unless stated otherwise, tables report the mean and sample standard deviation over the ten splits.

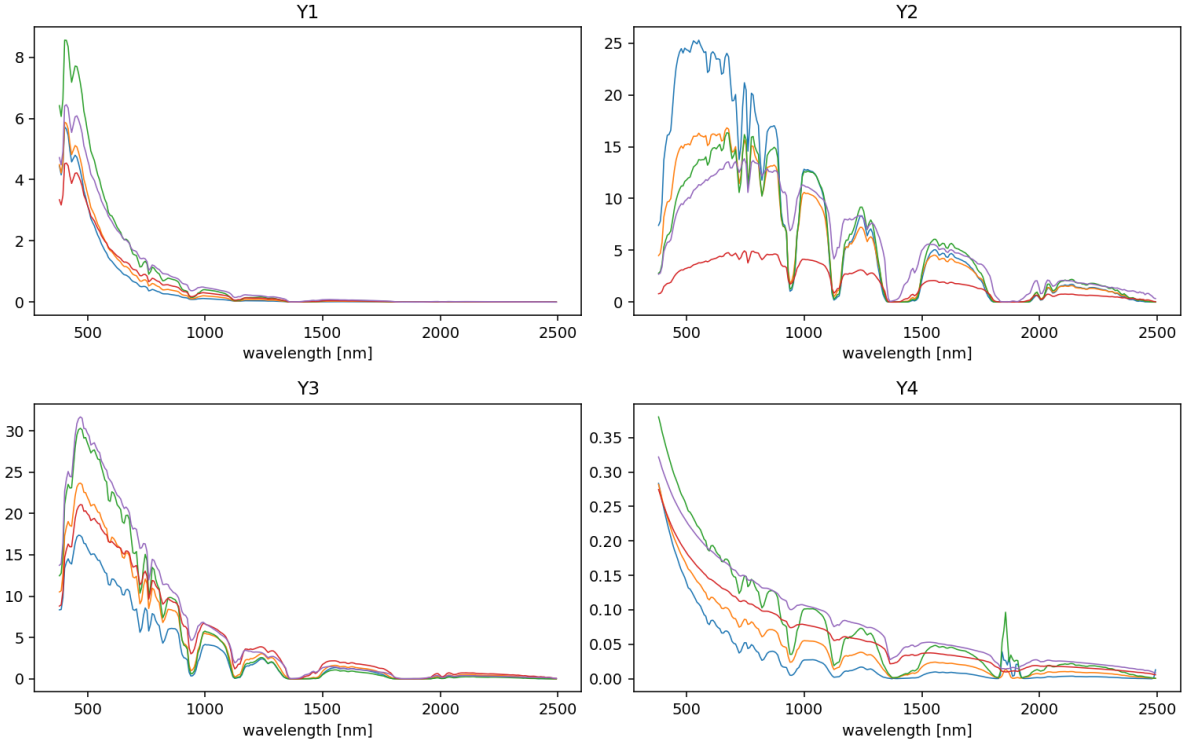


Figure 1: Example output curves across the EMIT wavelength grid, five samples per panel. The curves are smooth, share the deep water-vapor absorption features near 1380 and 1900 nm, and differ mainly by smooth rescalings along the state axes.

The model families and ablations were chosen during exploratory analysis of the same table, including inspection of test-set results. Repeated partitions therefore measure sensitivity to the split within this table, rather than independent external validation. Their test sets overlap, and their standard deviations are descriptive, not confidence intervals from ten independent datasets. Section 4.1 reports paired differences on common splits. Validation rows are shared by early stopping, tuning and model combination. Evaluation uses the single surface reflectance $\rho = 0.7$; other reflectances and atmospheric states outside the table are not tested.

Three empirical structural diagnostics motivate the model comparison.

The outputs are low rank. On standardized training outputs, 99.99% of the variance of every component lies in the first 11–25 principal components, and 99.9999% in at most 44 (Figure 2, left); the minimum adjacent-band correlation across the four components is 0.994952 on the training rows. All regressors below act on the first 64 principal coefficients, whose training unexplained-variance fraction is below 10^{-6} , except the networks, which are trained directly on the 285 standardized bands. A small aggregate variance tail is not a uniform relative-error guarantee in physical units; Section 6 states the projection error explicitly.

The measured input relevance is anisotropic. Shuffling the aerosol coordinate reduces a linear model’s validation R^2 by 0.89; the corresponding reduction is 0.54 for solar zenith and approximately 0.1 for elevation and water vapor. The two remaining angles have smaller effects (Figure 2, right). These validation diagnostics motivate input-specific kernel scales. They measure sensitivity of the fitted linear model, rather than identifying an intrinsic dimension of the

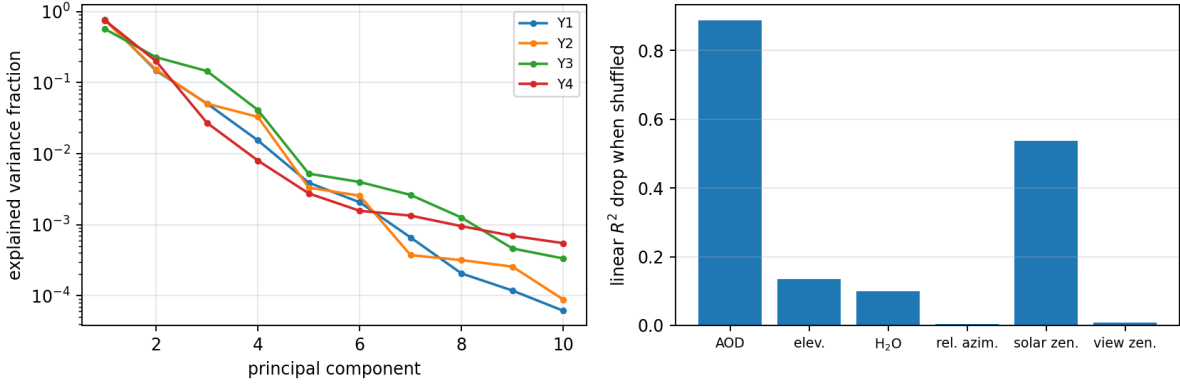


Figure 2: Left: explained-variance spectra of the standardized outputs. Every component concentrates 99.99% of its variance in 11–25 principal components out of 285 bands. Right: drop in linear validation R^2 when a single standardized input coordinate is shuffled. Aerosol optical depth and solar zenith angle dominate; the azimuth and view-zenith coordinates are almost inert at first order.

regression problem.

Low-degree surrogates are accurate. Linear regression on the standardized state achieves $R^2 = 0.83$, compared with 0.74 for distance-weighted five-nearest-neighbors. Cubic polynomial ridge regression has mean relative component error of approximately two percent (Table 1). Together with the spectral compression, these results motivate smooth low-dimensional regression as a baseline.

Reflectance inversion requires additional care. In deep absorption windows, the transmitted flux $Y_2 + Y_3$ can be as small as 10^{-21} . At a fully opaque band, the radiance is independent of surface reflectance and the inverse is non-identifiable. An auxiliary round-trip calculation on 2,332 states gives reflectance RMSE 0.0054 even with exact tabulated components. Its error is concentrated in 1,652 of 664,620 sample–band entries, whose inverse denominators are below 2×10^{-13} . All returned inverses are finite. This is a floating-point conditioning diagnostic, not a lower bound on emulator error. We report both the center and the tails of the reflectance-error distribution. Non-finite inverses are assigned zero by the scoring routine; their frequency is recorded separately.

Differentiating (2) separates the components. Write $D = \hat{Y}_2 + \hat{Y}_3 + \hat{Y}_4(L - \hat{Y}_1)$ for its denominator. Then

$$\frac{\partial \hat{\rho}}{\partial \hat{Y}_4} = -\hat{\rho}^2, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_2} = \frac{\partial \hat{\rho}}{\partial \hat{Y}_3} = -\frac{\hat{\rho}}{D}, \quad \frac{\partial \hat{\rho}}{\partial \hat{Y}_1} = -\frac{\hat{Y}_2 + \hat{Y}_3}{D^2}. \quad (3)$$

At exact identifiable components, $\hat{\rho} = \rho = 0.7$, so the albedo derivative is -0.49 , independently of the positive transmitted flux. For general predictions it equals $-\hat{\rho}^2$ and can be large. The denominator factors in (3) make path-radiance and flux errors particularly important near opaque bands. Section 4.1 reports the resulting all-band error tails.

2.1 Finite-perturbation control of the inversion

The derivatives in (3) describe local conditioning. The following elementary identities give a finite-error test and distinguish forward accuracy from inversion stability. For one sample and

band, write $a = Y_1$, $t = Y_2 + Y_3$, $s = Y_4$ and $L = a + \rho t / (1 - \rho s)$. Set $e_a = \hat{a} - a$, $e_t = \hat{t} - t$, $e_s = \hat{s} - s$.

Proposition 1 (Conditional radiance and reflectance stability). *Suppose $0 \leq \rho \leq \rho_{\max}$, $t > 0$ and $q = 1 - \rho s > 0$. If also $\hat{q} = 1 - \rho \hat{s} > 0$, the forward error satisfies the exact identity*

$$\hat{L} - L = e_a + \frac{\rho e_t}{\hat{q}} + \frac{\rho^2 t e_s}{q \hat{q}}. \quad (4)$$

In particular, if $q, \hat{q} \geq \kappa > 0$,

$$|\hat{L} - L| \leq |e_a| + \rho_{\max} |e_t| / \kappa + \rho_{\max}^2 |t e_s| / \kappa^2.$$

For inversion at the true radiance, put $u = L - a = \rho t / q$, $D = t / q$, and $\hat{D} = \hat{t} + \hat{s}(L - \hat{a})$. Whenever $\hat{D} \neq 0$,

$$\hat{\rho} - \rho = -\frac{(1 - \rho \hat{s})e_a + \rho e_t + \rho u e_s}{\hat{D}}, \quad \hat{D} - D = e_t + u e_s - \hat{s} e_a. \quad (5)$$

If $B = |e_t| + |u e_s| + |\hat{s} e_a| \leq \eta D$ for some $0 \leq \eta < 1$, then $\hat{D} \geq (1 - \eta)D > 0$ and

$$|\hat{\rho} - \rho| \leq \frac{|1 - \rho \hat{s}| |e_a| + \rho |e_t| + \rho |u e_s|}{(1 - \eta)D}. \quad (6)$$

Proof. Subtract the two forward maps and use $1/\hat{q} - 1/q = \rho e_s / (q \hat{q})$ to obtain (4). Since $u = \rho D$, $\hat{D} = D + e_t + u e_s - \hat{s} e_a$ and $\hat{\rho} - \rho = (u - e_a - \rho \hat{D}) / \hat{D}$, which gives (5). The triangle inequality yields $|\hat{D} - D| \leq B$ and then (6). $\square$

When $\hat{q} \neq 0$, the same calculation gives the exact forward-inverse link

$$\hat{\rho} - \rho = -\frac{\hat{q}}{\hat{D}}(\hat{L} - L). \quad (7)$$

Consequently, on a common set where $|\hat{q}| \leq Q$ and $|\hat{D}| \geq \delta > 0$, the corresponding unweighted L^2 errors obey $\|\hat{\rho} - \rho\|_2 \leq (Q/\delta)\|\hat{L} - L\|_2$. Without the denominator condition there is no uniform implication from small spectral relative radiance error to small reflectance error. For example, take two bands with $a = s = 0$, $t = (1, \varepsilon)$ and $0 < \rho < 1$. A predictor with $\hat{t} = t$, $\hat{s} = 0$ and $\hat{a} = (0, c\varepsilon)$, where $0 < c < \rho$, has relative spectral radiance error $c\varepsilon / (\rho\sqrt{1 + \varepsilon^2}) \rightarrow 0$ but retrieves $(\rho, \rho - c)$, so its reflectance RMSE remains $c/\sqrt{2}$. Every denominator is positive for $\varepsilon > 0$ and both retrieved reflectances remain physical. This example isolates the loss of inversion stability as the second-band flux vanishes.

The forward bound need not deteriorate when t tends to zero, whereas the inverse bound requires error control relative to $D = t/q$. Thus a small spectral radiance error does not certify stable reflectance recovery. At $t = 0$, $L = a$ is independent of ρ wherever the forward formula is defined, so reflectance is unidentifiable from this band. The identities are exact-arithmetic statements; floating-point cancellation in $L - a$ can further affect an implemented inverse.

For empirical comparison, a common application passband and conditioning threshold should be fixed before model evaluation. The utility `code/conditioned_reflectance.py` uses a positive flux scale determined on training data and a common truth-based mask, which is available in simulation benchmarks. It reports retained coverage, non-finite inversions and denominator failures separately. Masked errors require sample-level arrays and cannot be recovered from aggregate summaries. The EMIT comparisons in Section 4.1 therefore use the all-band statistics.

3 Models

All models are trained per component and evaluated in physical units after undoing the standardization; unless stated otherwise they regress the 64 leading principal coefficients of the standardized component, with the training variance criterion and projection qualification of Section 2. Every hyperparameter named below is selected on the validation set of the seed at hand. All experiments use CPU computation.

Reference family. Cubic polynomial ridge regression, acting on all monomials of degree at most three in the six standardized inputs, with the ridge weight selected from $\{10^{-10}, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. It provides a low-cost, low-degree reference; its empirical error alone does not prove a smoothness class for the underlying function.

Exact kernel ridge regression. Three constructions, all Matérn kernels on the standardized inputs with the length scale expressed as a multiple of the median pairwise distance of the fit set. The listed nugget is γ in $(K + n\gamma I)\alpha = Z$, not the unscaled diagonal term. The subset baseline is a Matérn-5/2 kernel fitted on a uniformly drawn subset of 4,000 training points, one eigendecomposition per length scale, multiplier in $\{1, 2\}$, nugget in $\{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$. The second is the exact solve on every training row: smoothness $\nu \in \{3/2, 5/2\}$, multiplier in $\{0.5, 1, 2, 4\}$ and the same nugget grid, tuned on validation from a 6,000-row training subsample and the winner refit on all 18,884 rows by one Cholesky factorization. The third gives each input its own length scale: starting from the isotropic winner, a coordinate search over per-input multipliers in $\{1/4, 1/2, 1, 2, 4\}$, two sweeps, scored on validation, after which the smoothness, scale and nugget are re-tuned at the found metric and the solve is refit on every row. The third construction allows the kernel to adapt to the input anisotropy measured in Section 2.

Fully connected networks. Three ELU hidden layers of 512 units per component, trained on the 285 standardized bands with AdamW at learning rate 10^{-3} , cosine decay, batch size 1024, weight decay 10^{-6} , up to 150 epochs with early stopping on validation loss (patience 25). Five networks are trained per component and seed from different initializations; the table reports the first and the mean of the five.

Kernel corrections of the neural mean. The construction of [5], with the exact kernel stages fitted on every training row. (i) *Residual correction*: an exact Matérn regression, on the standardized input space, of the network’s training residuals in coefficient space, added back to the network’s projection into that space, with smoothness, scale and nugget selected by the validation error of the corrected prediction; applied to the single network and to the five-network mean. (ii) *Kernel on features*: the same exact regression with the input metric replaced by the network’s last hidden layer, standardized, giving a kernel metric in the fitted feature representation. This is the post-hoc counterpart of deep kernel learning [22], in which the representation and the kernel are trained together, and of the kernel-flow regularization of a network’s inner layers [9]; here the representation is fixed first and the kernel is solved exactly on it. (iii) *Per-coordinate selection*: for each of the 64 output coefficients independently, pick on validation among the exact kernel, the network, the ensemble, the two corrected predictors and the feature kernel, and assemble the test prediction from the winners. (iv) *Convex stack*: per component, a convex combination of the same six heads with weights fitted on the validation

model	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [pp]
Cubic ridge	1.16±0.02	2.18±0.04	1.11±0.03	2.98±0.06	0.760±0.011	0.471±0.010
Matérn KRR, 4000-point fit	0.17±0.01	0.61±0.04	0.29±0.02	2.44±0.87	0.332±0.029	0.178±0.030
Matérn KRR, exact on all rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
Matérn KRR, per-input scales	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
FC-DNN (3×512)	0.48±0.02	0.97±0.04	0.41±0.02	3.02±0.18	0.387±0.012	0.245±0.008
Five-network mean	0.37±0.01	0.67±0.03	0.28±0.01	2.51±0.09	0.251±0.006	0.163±0.003
DNN + residual KRR	0.10±0.00	0.30±0.02	0.16±0.01	2.43±0.08	0.141±0.006	0.089±0.008
Ensemble + residual KRR	0.08±0.00	0.27±0.03	0.13±0.00	2.42±0.17	0.123±0.007	0.080±0.008
Kernel on network features	0.18±0.01	0.17±0.01	0.10±0.01	1.99±0.08	0.098±0.007	0.054±0.004
Per-coordinate selection	0.19±0.01	0.44±0.10	0.13±0.02	2.25±0.18	0.193±0.022	0.123±0.023
Convex stack	0.07±0.00	0.16±0.01	0.09±0.00	1.94±0.06	0.087±0.006	0.048±0.003

Table 1: Test errors under the protocol of Section 2, mean $\pm$ sample standard deviation over ten splits. Component and radiance columns report mean relative L^2 error; radiance is evaluated at $\rho = 0.7$. The last column is median absolute reflectance error in percentage points. The inverse is evaluated across all bands, including deep absorption windows, with the non-finite convention of Section 2. Table 3 reports the associated 95th-percentile errors and RMSE.

points in the physical relative metric, in the manner of stacked generalization [20] and stacked regression [21]. The choice of weighting rule for pools of this kind, and its relation to constrained minimum-variance weighting, is studied separately and is not part of this comparison.

Sample-size, rank and width experiments. Section 5 repeats the isotropic exact kernel, the network, its residual correction and the feature kernel at nested training sizes, at other output ranks and at a wider network; the table of Section 4 is the full-size, rank-64, width-512 cell of that grid.

4 Results

Table 1 compares the model families at 18,884 training rows, output rank 64 and network width 512.

Polynomial and subset-kernel baselines. Cubic ridge regression gives 1.86% mean component error and 0.760% radiance error. The 4,000-point Matérn fit reduces these means to 0.88% and 0.332%. Its albedo error varies substantially across splits: approximately 1.9% on seven splits and 3.3–4.3% on the remaining three.

Full-data and input-scaled kernels. The isotropic full-data kernel reduces mean radiance error to 0.239% and path-radiance error to 0.08%, compared with 0.17% for the subset fit. Its albedo error is 2.3–2.9% across splits: less variable, but larger than the subset fit on the latter’s seven most accurate splits. Input-specific length scales give 0.095% radiance error, 0.63% mean component error and 0.036 percentage points of median reflectance error. The fitted scales are consistent with weak azimuth dependence of the flux components. The relative-azimuth length scale is sixteen to thirty-two times larger than the other scales, with median factors sixteen for direct flux and albedo and thirty-two for diffuse flux. The longer azimuth scale occurs on every

split for the flux components and nine of ten for albedo. View-zenith scales are two to sixteen times longer, whereas the path-radiance metric is approximately isotropic.

Networks and kernel corrections. The network has 1.22% mean component error and 0.387% radiance error. Averaging five initializations reduces these to 0.96% and 0.251%. Residual kernel regression reduces the single-network radiance error to 0.141% and the ensemble error to 0.123%. The feature kernel has 0.61% mean component error and 0.098% radiance error. Its radiance comparison with the input-scaled kernel is close: the feature kernel is lower on six of ten splits, although its mean is slightly larger. Its median reflectance error is 0.054 points, compared with 0.036 for the input-scaled kernel, and is larger on every split. The separately tuned residual correction substantially improves the network, but has larger radiance error than the input-scaled and feature kernels.

Selection and stacking. Both combiners use the isotropic full-data kernel, the network, its five-member ensemble, the two residual-corrected predictors and the feature kernel. The input-scaled kernel is not in this pool. Coordinatewise validation selection gives 0.193% radiance error, larger than several individual members. The convex stack gives 0.087% radiance error, 0.57% mean component error and 0.048 points of median reflectance error. Its radiance gain is approximately 8% relative to the input-scaled kernel and 11% relative to its best individual member, the feature kernel. On mean component error, the stack improves on the input-scaled kernel at nine splits, with paired difference 0.06 ± 0.05 points, and on the feature kernel at all ten, with difference 0.047 ± 0.006 points.

The residual second moments provide a diagnostic of complementarity. On path radiance, the isotropic kernel and five-network ensemble have residual correlation 0.27 ± 0.09 , below their error ratio of approximately 0.34 at every split. For a quadratic error metric, this is the classical condition under which a convex combination can improve on the better member [23, 5]. The stack instead minimizes a mean relative norm, and its reflectance objective is nonlinear; the diagnostic does not guarantee improvement in those metrics. The two residual-corrected predictors have much larger correlations, approximately 0.88 on path radiance and 0.99 on albedo. Coordinatewise selection also differs from stacking in its objective and number of independently chosen coefficients.

Spherical albedo. Albedo has the largest relative error for every family. Its small short-wave-infrared values make relative errors sensitive to absolute prediction errors in that region. The width-512 stack achieves 1.94% albedo error. The learning curves below show little improvement with sample size on this component; the experiments do not isolate whether the plateau arises from approximation, optimization or the relative-error weighting.

4.1 Paired comparisons and the radiance–reflectance trade-off

Paired differences compare candidate and reference errors on the same split. Table 2 reports those differences for the main and wide pipelines. Positive gains favor the candidate. The wide feature kernel improves on the input-scaled kernel at all ten splits, by 0.0165 ± 0.0066 radiance percentage points. By contrast, the wide stack’s improvement over its own feature kernel is only 0.0006 ± 0.0026 points, with eight wins and two losses. The mean improvement is small relative to the variability of the paired differences. These are descriptive paired summaries: repeated

candidate / reference	mean gain $\pm$ SD [pp]	gain range [pp]	wins/10
Input-scaled / isotropic KRR	0.1436 $\pm$ 0.0122	[0.1196, 0.1585]	10
Residual correction / network	0.2464 $\pm$ 0.0090	[0.2365, 0.2615]	10
Narrow stack / input-scaled KRR	0.0083 $\pm$ 0.0066	[-0.0054, 0.0185]	9
Wide feature kernel / input-scaled KRR	0.0165 $\pm$ 0.0066	[0.0054, 0.0295]	10
Wide / narrow feature kernel	0.0191 $\pm$ 0.0079	[0.0096, 0.0352]	10
Wide stack / wide feature kernel	0.0006 $\pm$ 0.0026	[-0.0053, 0.0037]	8

Table 2: Paired radiance contrasts on seeds 101–110. Gain is the reference error minus the candidate error, in percentage points (pp). SD is the sample standard deviation of the ten paired differences; the range and win count describe those same ten differences. They are not confidence intervals or tests of a population-level superiority claim.

model	radiance [%]	median [pp]	95th percentile [pp]	RMSE [pp]
Cubic ridge	0.760 $\pm$ 0.011	0.471 $\pm$ 0.010	70.68 $\pm$ 0.02	4.18e+03
Isotropic KRR	0.239 $\pm$ 0.016	0.141 $\pm$ 0.018	69.53 $\pm$ 0.42	4.18e+03
Input-scaled KRR	0.095 $\pm$ 0.009	0.036 $\pm$ 0.004	31.16 $\pm$ 6.21	1.34e+04
Narrow feature kernel	0.098 $\pm$ 0.007	0.054 $\pm$ 0.004	7.75 $\pm$ 1.13	1.78e+03
Narrow convex stack	0.087 $\pm$ 0.006	0.048 $\pm$ 0.003	29.44 $\pm$ 5.89	1.25e+03
Wide feature kernel	0.079 $\pm$ 0.009	0.045 $\pm$ 0.006	7.57 $\pm$ 0.95	1.40e+03
Wide convex stack	0.078 $\pm$ 0.011	0.043 $\pm$ 0.005	22.71 $\pm$ 4.42	5.81e+03

Table 3: All-band metrics from the complete ten-seed main and wide campaign records. Radiance is mean relative L^2 error. Reflectance median, 95th-percentile absolute error and RMSE are in percentage points at $\rho = 0.7$. The first three columns show mean $\pm$ sample SD across splits; RMSE shows its mean only, with its spread retained in the machine-readable summary. The inverse is neither physically clipped nor conditioned on an application passband. All inverses in these records are finite, so their zero-fill branch does not explain the large errors. Large finite values near denominator zeros dominate RMSE. These are diagnostic all-band errors, not operational retrieval accuracy.

splits of one finite table, overlapping test rows and adaptive study development do not supply ten independent external validation experiments.

Radiance and reflectance also order the heads differently. Table 3 reports the all-band reflectance tails alongside the medians. Each entry averages a within-split statistic; the mean of ten 95th percentiles is not a pooled 95th percentile. For the width-512 pipeline, the convex stack lowers radiance error relative to the feature kernel, but raises the average 95th-percentile reflectance error from 7.75 to 29.44 percentage points. The tail is worse at all ten splits, by factors between 2.63 and 4.94. The width-2000 comparison repeats the reversal: the feature kernel has 7.57 points at the 95th percentile and the stack 22.71, with the stack worse at all ten splits by factors between 2.36 and 3.79. Thus the near-tie at 0.079% versus 0.078% radiance error conceals a substantial difference in the inverse metric.

The input-scaled kernel illustrates the same distinction: its median error is only 0.0365 points but its mean 95th percentile is 31.16 points. Conversely, the feature kernels have slightly larger medians and appreciably smaller tails. Model rankings depend on the metric and spectral

training rows	seeds	Cubic ridge	KRR, isotropic	KRR, per-input	FC-DNN	DNN + residual KRR	kernel on features
500	10	0.846±0.031	0.639±0.022	0.415±0.024	1.961±0.057	0.815±0.035	0.456±0.020
1,000	10	0.802±0.024	0.519±0.034	0.260±0.066	1.862±0.015	0.590±0.027	0.316±0.040
2,000	10	0.780±0.016	0.424±0.051	0.217±0.052	1.232±0.034	0.376±0.041	0.240±0.049
4,000	10	0.769±0.010	0.328±0.028	0.154±0.037	0.830±0.019	0.256±0.021	0.164±0.033
8,000	10	0.763±0.010	0.279±0.020	0.119±0.027	0.584±0.020	0.190±0.014	0.121±0.017
18,884	10	0.760±0.011	0.239±0.016	0.095±0.009	0.387±0.012	0.141±0.006	0.098±0.007

Table 4: Learning curves: radiance relative L^2 error [%] against the number of training rows, mean $\pm$ standard deviation over the ten splits reported in the archived aggregate. At $n = 8,000$, one individual record per kernel campaign is absent from the public archive; the table reports the ten-run aggregate. Each rung uses the first n rows of the seed’s training block; the validation and test blocks are those of Table 1, so the top rung is that table. The per-input kernel repeats its coordinate search at every rung. Component columns are in Figure 3 and in the per-seed records.

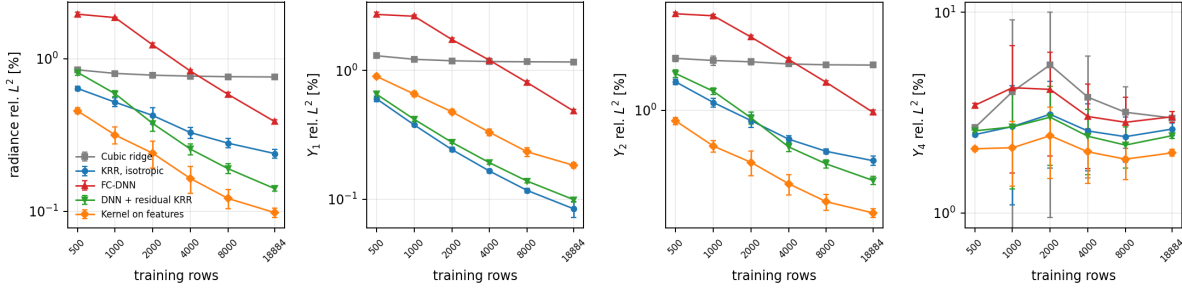


Figure 3: Learning curves on log–log axes for the radiance and for three of the four components, mean and standard deviation over ten splits. Cubic error changes little over the sample-size range. The kernel, network, residual-corrected network and feature kernel decrease approximately as $n^{-1/2}$ on the first three components, with fitted slopes in Table 5. Albedo error shows no sustained decrease; unusually large errors on individual splits increase its spread at 1,000 and 2,000 rows.

domain. The identities of Section 2.1 relate inverse error to the predicted denominator as well as forward error. This explains why optimizing componentwise or radiance error alone need not control reflectance tails; it does not identify the component-error correlations responsible for a particular observed tail.

The tables are generated from the twenty individual records. The generator checks split configurations, agreement of recorded array digests, and equality of common deterministic base-lines. These checks establish consistency of the summaries; reproducing model training requires the raw arrays.

5 Learning curves, output rank and network width

The following experiments vary training size, output rank and network width, using the same validation-based selection. Learning curves include cubic ridge, isotropic and input-scaled kernels, the network, residual correction and the feature kernel. Ensemble comparisons use the full training size. The number of available repetitions is given with each experiment.

family	slope over the six rungs					slope of the last step				
	Y_1	Y_2	Y_3	Y_4	radiance	Y_1	Y_2	Y_3	Y_4	radiance
Cubic ridge	-0.03	-0.03	-0.03	-0.02	-0.03	-0.00	-0.01	-0.00	-0.08	-0.01
KRR, isotropic	-0.55	-0.38	-0.40	-0.01	-0.28	-0.38	-0.19	-0.17	0.10	-0.18
KRR, per-input	-0.45	-0.43	-0.43	0.01	-0.40	-0.32	-0.26	-0.29	0.12	-0.26
FC-DNN	-0.50	-0.49	-0.57	-0.09	-0.48	-0.59	-0.60	-0.55	0.07	-0.48
DNN + residual KRR	-0.52	-0.53	-0.61	-0.04	-0.50	-0.39	-0.33	-0.39	0.12	-0.35
Kernel on features	-0.45	-0.44	-0.46	-0.03	-0.43	-0.28	-0.22	-0.31	0.08	-0.25

Table 5: Learning-curve exponents: the least-squares slope of log error against log n over the six rungs of Table 4, and the slope of the last step alone, from 8,000 to 18,884 rows, per family and component.

Learning curves. Each rung uses the first n rows of the seed’s training block, $n = 500, 1,000, 2,000, 4,000, 8,000$ and the full 18,884, with the validation and test blocks unchanged, so the rungs of one seed are nested and the full block is the top rung (Table 4, Figure 3). Cubic error changes little: from 0.85% to 0.76% over a factor of 38 in data, with most of its decrease achieved by a thousand rows. On the first three components every other family decays at close to $n^{-1/2}$ (Table 5): the least-squares slopes of log error against log n over the six rungs lie between -0.38 and -0.61 for the kernel, the network, the corrected network and the feature kernel on Y_1 , Y_2 and Y_3 , while on the albedo every slope lies between -0.09 and $+0.01$: the albedo shows no sustained decrease over this budget range. This plateau does not distinguish approximation bias, optimization, relative-error weighting or limitations of the training data.

These are slopes of six mean errors over a finite sample-size range, not established asymptotic rates. Standard Sobolev or kernel regression rate formulas require sampling, source and regularization assumptions [13, 15]; neither those assumptions nor the smoothness norm of the fitted network residual is verified by this experiment. The isotropic kernel’s decrease is slower at the largest sample size. Its slope over the first five rungs is -0.59 , -0.44 and -0.47 on the smooth components, and its last step, from 8,000 to 18,884 rows, is -0.38 , -0.19 and -0.17 , while the network’s last step is as steep as its average (-0.59 , -0.60 and -0.55 ; Table 5). Part of this is the tuning protocol: the kernel’s smoothness, scale and nugget are chosen on a 6,000-row subsample at every rung, so above 6,000 rows the hyperparameters are those of a smaller problem, and Section 5.1 measures what tuning with every row in the solve recovers: a substantial part of the measured change in slope. The consequence for the comparison is that the ratio of the network’s radiance error to the kernel’s falls from 3.1 at 500 rows to 1.6 at the full block, or to 1.8 with the kernel tuned on every row. The kernel’s margin over the network on this table is a margin at this training size; whether it would survive ten times the data depends on where the kernel’s curve settles, which the bend leaves open.

Relative to the isotropic kernel, the per-input metric changes the measured finite-range slope as well as the error level. Its coordinate search, repeated at every rung, gives a slope of -0.399 on the radiance against the isotropic kernel’s -0.279 , and the ratio between them falls from 0.65 at 500 rows to 0.40 at the full block, with one step against that direction (0.50 at 1,000 rows, 0.51 at 2,000) and a fall at every other. The relative error reduction increases over this sample-size range. The selected metric, smoothness and nugget all change with sample size, so these measurements do not isolate a change of asymptotic dimension or establish a faster universal rate. Against the kernel on network features the margin closes as the data grows. The per-input metric has the lower mean at all six rungs, but the split-by-split count runs 10, 9, 10, 7, 5 and

PCA rank	variance outside the basis	KRR, isotropic	FC-DNN	DNN + residual KRR	kernel on features
16	7.5×10^{-4}	0.270 ± 0.015	0.390 ± 0.015	0.175 ± 0.007	0.148 ± 0.009
32	1.5×10^{-5}	0.243 ± 0.018	0.390 ± 0.015	0.139 ± 0.006	0.102 ± 0.009
64	8.2×10^{-8}	0.242 ± 0.018	0.390 ± 0.015	0.143 ± 0.006	0.100 ± 0.009
128	5.4×10^{-9}	0.242 ± 0.018	0.390 ± 0.015	0.150 ± 0.005	0.100 ± 0.009
285	0	0.242 ± 0.018	0.390 ± 0.015	0.172 ± 0.004	0.100 ± 0.009

Table 6: Output-rank ablation: radiance relative L^2 error [%] when the kernels and corrections regress the leading r principal coefficients, mean $\pm$ standard deviation over five splits. The second column is the fraction of the standardized variance the basis leaves out for the slowest-decaying component, the spherical albedo; at rank 285 it is zero. The network regresses the 285 bands directly and is the same predictor in every row.

6 wins of ten from 500 rows to the full block, and the gap itself narrows from 0.415 against 0.456 at 500 rows to 0.095 against 0.098 at the top; the feature kernel’s own slope, -0.435 , is the steeper of the two. The input-scaled kernel has a larger advantage at smaller sample sizes, while the feature kernel approaches it as the training set grows. On the albedo the per-input slope is $+0.01$: with no sustained improvement over this range.

The corrected network is below both of its parents on the radiance at every one of the ten splits from 2,000 rows on, and at no split at 500 or 1,000 rows, where its error sits between the network’s and the kernel’s. On Y_2 and Y_3 it is below both at every split at the two largest sizes; on Y_1 it never is, the isotropic kernel alone being the better head at every size (0.084% against 0.099% at the full block). Its exponent is close to the network’s and its error level is closer to the kernel’s in these runs. Section 6 gives the exact residual decomposition, while distinguishing it from a proof of these empirical slopes or of the component-specific exception.

Output rank. The kernels and corrections regress the leading r principal coefficients of the standardized component; the network regresses the 285 bands and is the same predictor at every rank. Table 6 refits at $r = 16, 32, 128$ and 285 at five of the seeds beside the rank-64 rows of those seeds. The kernel’s measured error changes little above rank 32: its radiance error reads 0.243, 0.242, 0.242 and 0.242 at 32, 64, 128 and 285, and 0.270 at 16, where the truncation leaves 7.5×10^{-4} of the albedo’s variance outside the basis. The kernel on features behaves the same way. The residual correction does not: it is best at 32 (0.139), a little worse at 64 (0.143), and worse again at 128 and 285 (0.150 and 0.172). The two regressions fit different targets: the direct kernel fits the component, whereas the correction fits the projected network residual. Increasing rank therefore exposes the correction to additional network-error directions, even when the target variance in those directions is small. The lowest measured correction errors occur at ranks 32–64. The projection identity in Section 6 separates these two error terms.

A wider network. A network of three hidden layers of 2,000 units, trained for up to 500 epochs under the same optimizer, learning rate and early stopping as the 512-unit network, was run at all ten seeds with the rest of the pipeline built on it (Table 7). It is a worse emulator at every seed, 0.75% against 0.39% on the radiance, and its early stopping fired between 42 and 210 epochs of the 500 allowed: at this learning rate and batch size this particular wider-network training configuration gives worse standalone predictions. Width and epoch budget change together; these results do not rule out a better-optimized wide network. The exact kernel on its last-layer features reaches 0.079% on the radiance against 0.098% on the 512-unit features,

network	head	component rel. L^2 [%]				radiance	reflectance
		Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [pp]
3×512	network	0.48 ± 0.02	0.97 ± 0.04	0.41 ± 0.02	3.02 ± 0.18	0.387 ± 0.012	0.245 ± 0.008
	network + residual KRR	0.10 ± 0.00	0.30 ± 0.02	0.16 ± 0.01	2.43 ± 0.08	0.141 ± 0.006	0.089 ± 0.008
	kernel on its features	0.18 ± 0.01	0.17 ± 0.01	0.10 ± 0.01	1.99 ± 0.08	0.098 ± 0.007	0.054 ± 0.004
	per-coordinate selection	0.19 ± 0.01	0.44 ± 0.10	0.13 ± 0.02	2.25 ± 0.18	0.193 ± 0.022	0.123 ± 0.023
	convex stack	0.07 ± 0.00	0.16 ± 0.01	0.09 ± 0.00	1.94 ± 0.06	0.087 ± 0.006	0.048 ± 0.003
3×2000	network	0.93 ± 0.20	2.11 ± 0.49	0.76 ± 0.19	3.27 ± 0.19	0.748 ± 0.116	0.508 ± 0.110
	network + residual KRR	0.12 ± 0.01	0.35 ± 0.03	0.19 ± 0.03	2.35 ± 0.13	0.160 ± 0.016	0.106 ± 0.010
	kernel on its features	0.13 ± 0.02	0.13 ± 0.02	0.08 ± 0.01	2.10 ± 0.34	0.079 ± 0.009	0.045 ± 0.006
	per-coordinate selection	0.14 ± 0.02	0.30 ± 0.26	0.09 ± 0.01	2.55 ± 0.26	0.187 ± 0.057	0.099 ± 0.033
	convex stack	0.08 ± 0.01	0.13 ± 0.02	0.08 ± 0.00	2.00 ± 0.24	0.078 ± 0.011	0.043 ± 0.005

Table 7: The pipeline of Section 3 run with a 3×2000 network in place of the 3×512 one, at the same ten splits: the network (one member, up to 500 epochs under the same optimizer, learning rate and early stopping), its residual correction, the exact kernel on its last-layer features, and the per-coordinate selection and convex stack over the wide pipeline’s heads (the isotropic kernel, the network, its correction and its feature kernel), beside the corresponding rows of Table 1; mean $\pm$ standard deviation over the ten splits.

below the per-input kernel’s 0.095% at all ten seeds and below the convex stack of Table 1 at nine of them, and on the two flux components, at 0.13% and 0.08%, it is at or below every row of that table. The convex stack over the wide pipeline’s heads, which puts 0.88 to 0.96 of its weight on that feature kernel, has the lowest mean radiance error in the ten-split comparison at 0.078% on the radiance and 0.043 reflectance points, below the stack of Table 1 at nine of the ten seeds; the per-coordinate selection over the same heads does not transfer here either. The residual correction of the wider network is a little worse than that of the narrower one, although that ordering is not guaranteed merely because its base network is worse (Section 6). A 3×1000 rung at the same ten seeds sits between the two: the network at 0.72%, the kernel on its features at 0.088% against 0.098% at width 512 and 0.079% at width 2,000. From 512 to 2,000 the network’s own error rises with width while the error of a kernel on its features falls: the fitted features improve kernel prediction despite the larger standalone network error. At 4,000 units, run at two seeds, the network is worse again at 0.95% and the kernel on its features turns back up to 0.097%, so the gain in the representation does not continue past 2,000.

A three-member wide-network experiment, run at three seeds, supplies the exact kernel with the concatenated last-layer features of all three members at once, 6,000 coordinates rather than 2,000. It reads 0.074% on the radiance against 0.083% for the kernel on a single member’s features on the same splits, lower at all three seeds, and 0.043 against 0.051 reflectance points, again at all three. That puts one exact solve level on the radiance with the convex stack of six heads fitted on those same splits (0.074% against 0.074% on the mean, though the stack is the lower at two of the three seeds and on the reflectance median at all three). The six-predictor stack and the single kernel on concatenated features have similar mean radiance errors on these three splits. The comparison does not isolate all optimization and kernel-selection effects. Three seeds provide exploratory evidence rather than a ten-split replication, and the concatenated-feature construction was evaluated only at width 2,000.

kernel	component rel. L^2 [%]				radiance	reflectance
	Y_1	Y_2	Y_3	Y_4	rel. L^2 [%]	med. $ \hat{\rho} - \rho $ [%]
isotropic, tuned on 6,000 rows	0.08±0.01	0.42±0.03	0.20±0.00	2.62±0.18	0.239±0.016	0.141±0.018
isotropic, tuned on every row	0.08±0.00	0.36±0.02	0.17±0.01	2.58±0.14	0.210±0.008	0.107±0.006
per-input search, tuned on 6,000 rows	0.08±0.01	0.16±0.02	0.08±0.01	2.18±0.16	0.095±0.009	0.036±0.004
per-input search, re-tuned on every row	0.08±0.00	0.16±0.02	0.08±0.01	2.09±0.12	0.096±0.009	0.036±0.004
per-input from the cubic’s gradients, $p = 1$	0.34±0.01	0.19±0.02	0.14±0.01	2.17±0.14	0.159±0.007	0.078±0.004
per-input from the cubic’s gradients, $p = 1/2$	0.11±0.00	0.19±0.02	0.09±0.01	2.27±0.15	0.125±0.008	0.055±0.004
per-input from the cubic’s gradients, $p = 1/4$	0.08±0.00	0.25±0.02	0.14±0.01	2.38±0.20	0.157±0.015	0.077±0.013

Table 8: Where the kernel’s hyperparameters are chosen, and how its metric is obtained, at the full training block and the same ten splits. The first pair tunes smoothness, scale and nugget on a 6,000-row subsample and on every training row; the last three rows replace the coordinate search by a metric read off the cubic surrogate’s gradients, at three exponents. Selecting the exponent on validation, per seed, chooses $p = 1/2$ at all ten, so that row would duplicate the $p = 1/2$ row and is not printed; the per-seed choices are in the stored numbers. Mean $\pm$ standard deviation over the ten splits.

5.1 Full-data tuning and gradient-based metrics

We compare hyperparameter selection on a 6,000-row subsample with selection on all training rows, and compare coordinate search with a metric derived from a cubic surrogate’s gradients (Table 8). The reported aggregates contain ten splits; the per-split archive for full-data tuning contains five of them.

Tuning with every row in the solve. The kernel families of Section 3 choose smoothness, scale and nugget by the validation error of a solve on a 6,000-row subsample, and refit the winner on all 18,884 rows. Doing the tuning solve on every row instead lowers the isotropic kernel’s radiance error from 0.239% to 0.210% and its reflectance median from 0.141 to 0.107 points, better at nine of the ten splits, the ratio averaging 0.882 and never above 1.001. The selected scales differ: over the forty component-seed cells the full block picks a shorter length scale than the subsample in 26, the same in 12, and a longer one in 2. In this experiment, subsample selection usually chooses a longer length scale than full-data selection, consistent with the smaller tuning fit favoring greater smoothing.

Below 6,000 rows the tuning subsample *is* the whole training block and the rungs are already full-block tuned; at 8,000 rows it is three quarters of it; only the top rung is a third. Applying the measured ratio to the top rung of Table 4 takes the kernel’s last step on the radiance from -0.183 to -0.334 , against a slope of -0.305 over the five lower rungs: the top-step slope becomes closer to that of the lower rungs. One retuned endpoint does not establish a power law over or beyond the observed range. On the two flux components the same correction takes the last step from -0.187 and -0.171 to -0.352 and -0.358 against averages of -0.437 and -0.468 , reducing, but not eliminating, the difference between the last-step and lower-rung slopes. The consequence for the comparison of Section 4 is that the kernel’s margin over the network at the full block is wider than Table 4 shows: 1.8 times rather than 1.6, under this retuned endpoint. No conclusion about unobserved sample sizes follows.

Full-data tuning has little effect on the input-scaled kernel: 0.096% against 0.095%, better at four of the ten splits, ratio 1.005. Its coordinate search already selects a scale of once the median distance at the metric it finds, which is what the full-block tune selects too. The input-scaled kernel is therefore less sensitive to the tuning sample size in these experiments.

Gradient-based metric. The coordinate search costs a sweep over the six inputs at five multipliers, twice, per component and per seed. The alternative sets the metric before any kernel is fitted: the multiplier of coordinate j is the root-mean-square gradient of the fitted cubic surrogate along j over the training rows, raised to a power p , normalized to geometric mean one and clipped to the search’s own range $[1/16, 4]$, so weakly varying coordinates receive longer length scales. The exponent p controls the strength of the rescaling. It costs one cubic fit and six finite differences.

At $p = 1$, radiance error is 0.159%, below the isotropic kernel at every one of the ten splits by a factor 0.669 on average and above the searched metric at every one of them by a factor 1.680. The dependence on p is nonmonotone: $p = 1/2$ gives 0.125% and $p = 1/4$ gives 0.157%, so the best exponent is interior, and choosing it the way every other hyperparameter in this study is chosen — the lowest mean validation error over the four components, per seed, with the test block untouched — selects $p = 1/2$ at all ten seeds. At that exponent the metric closes 79% of the distance from the isotropic kernel to the searched one on the radiance and 82% on the reflectance median, for one cubic fit, six finite differences and a choice among three numbers.

The componentwise errors favor different exponents: Y_2 and Y_3 are best at $1/2$, the albedo at 1 (2.18% against 2.27%), and the path radiance at $1/4$. On that component the isotropic kernel already reads 0.084% and the searched metric 0.085%: coordinate search gives little improvement. The proportional metric nonetheless reads 0.34% there, approximately four times the isotropic error. A possible explanation is that the surrogate’s gradient along a coordinate measures how fast the map moves with it and not how much predictive information it carries, so gradient magnitude alone need not provide an appropriate kernel metric. Softening the exponent to $1/4$ takes Y_1 back to 0.085%, exactly the searched metric’s value, at the cost of Y_2 and Y_3 . The validated gradient metric recovers much of the observed radiance gain with fewer metric searches. Coordinate search additionally allows independent componentwise adjustments, whereas the gradient construction selects one exponent for all four components.

6 Residual correction and error alignment

Fix the training design, neural mean and selected kernel hyperparameters. The following identities hold without sampling assumptions. We first work with one output coordinate; the vector-valued statements follow coordinatewise.

6.1 A frozen regression operator

Fix a design $X_n = (x_1, \dots, x_n)$, a positive-semidefinite kernel with Gram matrix K , and $\lambda > 0$. Define

$$(Sg)(x) = k(x, X_n)(K + \lambda I)^{-1}g(X_n), \quad R = I - S. \quad (8)$$

Here λ is the diagonal regularizer in the linear system. The campaign code uses $\lambda = n\gamma$, where γ is its reported nugget; the distinction matters when the training size changes. Although a selected kernel can depend on the targets, S is linear in the response only *after that selection is frozen*.

Proposition 2 (Frozen residual correction). *Let f be the target and h any fitted mean, and set*

$\hat{f}_c = h + S(f - h)$. Then:

$$f - \hat{f}_c = R(f - h), \quad (9)$$

$$(Rg)(X_n) = \lambda(K + \lambda I)^{-1}g(X_n). \quad (10)$$

If $Ku_j = \mu_j u_j$, the design residual in direction u_j is multiplied by $\lambda/(\mu_j + \lambda) \in (0, 1]$. The factor is one when $\mu_j = 0$ and strictly below one when $\mu_j > 0$; this is shrinkage, not a hard spectral projection.

Suppose, in addition, that $\mathcal{S}$ is a normed vector space containing f and h and that the following bound holds uniformly on that space for the realized operator:

$$\|Rg\|_{L^2(P)} \leq C_n \|g\|_{\mathcal{S}} \quad (g \in \mathcal{S}).$$

Then

$$\|f - \hat{f}_c\|_{L^2(P)} \leq C_n \|f - h\|_{\mathcal{S}}. \quad (11)$$

Consequently, $\|f - h\|_{\mathcal{S}} < \|f\|_{\mathcal{S}}$ makes this displayed upper bound smaller than $C_n \|f\|_{\mathcal{S}}$ when $C_n > 0$. It is neither a necessary nor a sufficient condition for the corrected predictor's actual error to be smaller than that of Sf .

Proof. Subtract $h + S(f - h)$ from f to obtain (9). On the design, $I - K(K + \lambda I)^{-1} = \lambda(K + \lambda I)^{-1}$, which proves (10) and its spectral form. Apply the assumed uniform inequality to $g = f - h$ for (11). The examples below distinguish this bound from actual error comparisons. $\square$

Why the bounds do not order the errors. On a two-point design take $K = \text{diag}(1, 99)$ and $\lambda = 1$, so $R = \text{diag}(1/2, 1/100)$ and $C_n = 1/2$ is its Euclidean operator norm. With $f = (1, 0)$ and $f - h = (0, 2)$, the residual norm is twice the target norm, yet $\|R(f - h)\| = 0.02 < 0.50 = \|Rf\|$. Conversely, with $f = (0, 1)$ and $f - h = (1/2, 0)$, the residual norm is smaller, but $\|R(f - h)\| = 0.25 > 0.01 = \|Rf\|$. These examples also show why source-norm accuracy alone does not order corrected prediction errors.

6.2 An exact criterion for improvement

The relevant comparison can be made exactly in any Hilbert error metric. This includes a finite weighted squared error and $L^2(P)$, but not directly the mean of per-sample relative norms or the nonlinear radiance and reflectance metrics in Table 1.

Proposition 3 (Error alignment). *For the same frozen operator S , put $a = Rf$ and $b = Rh$. Then*

$$\|f - \hat{f}_c\|^2 - \|f - Sf\|^2 = \|b\|^2 - 2\langle a, b \rangle. \quad (12)$$

Thus full correction improves on Sf exactly when $2\langle a, b \rangle > \|b\|^2$. More generally, the interpolation $p_t = Sf + tRh$, $0 \leq t \leq 1$, has minimum squared error at

$$t_* = \min\{1, \max\{0, \langle a, b \rangle / \|b\|^2\}\} \quad \text{if } b \neq 0; \quad (13)$$

if $b = 0$, every t is equivalent. If the direct fit uses S_0 and the correction uses S_1 , (12) instead holds with $a = (I - S_0)f$ and $b = (S_1 - S_0)f + (I - S_1)h$.

Proof. For a common operator the corrected error is $a - b$, so expansion of its squared norm gives (12). The error of p_t is $a - tb$; minimize the resulting quadratic on $[0, 1]$ to obtain (13). For distinct operators, $f - h - S_1(f - h) = (I - S_0)f - [(S_1 - S_0)f + (I - S_1)h]$, after which the same expansion applies. $\square$

The criterion depends on the unknown target error. Replacing its inner products by validation errors gives a validation-optimal interpolation, not a guarantee on new data. The reported experiments use separately tuned direct and residual kernels, so their comparison must use the final, two-operator form with their actual selected operators. Neither the criterion nor the training-design contraction establishes that correction will beat both parents at every sample size.

6.3 What the principal-component implementation computes

The campaign trains a network on all standardized bands but projects its predictions into the retained principal-component basis before correction. Thus its corrected predictor is based on the projected network mean. To state the implementation precisely, let f and h now denote the target and network output after the common fitted standardization and PCA centering. Let Π be the orthogonal projection onto the retained spectral coordinates. In these centered coordinates, the implemented prediction is

$$\hat{f}_{c,\Pi} = \Pi h + S\Pi(f - h).$$

The output mean and diagonal band scaling are restored afterwards. Since the same scalar kernel operator acts on each retained output coordinate,

$$f - \hat{f}_{c,\Pi} = (I - \Pi)f + (I - S)\Pi(f - h). \quad (14)$$

The terms on the right are orthogonal in the standardized spectral Euclidean norm at each query, hence their squared norms add there. They need not be orthogonal in physical units after unequal band rescaling. This decomposition separates truncation error from residual regression and explains why output rank is a genuine part of the model specification. It does not by itself prove the observed optimum near rank 32. Comparisons with the unprojected network include the effect of projection as well as residual fitting.

Conditional rate bound. Kernel learning-rate results impose source, spectral, sampling and regularization conditions [13, 14, 15]. If a uniform residual bound holds with $C_n = O(n^{-\alpha})$ and $\|f - h_n\|_S = O(n^{-\beta})$, then (11) gives $\|f - \hat{f}_c\|_{L^2(P)} = O(n^{-(\alpha+\beta)})$. The source-norm bound and the uniformity required for a data-dependent residual are additional hypotheses. They do not follow from empirical network loss. The experiments measure finite-range error slopes, not these source norms or an asymptotic rate.

The code checks the residual identities, regularizer convention, semidefinite boundary, alignment formula, counterexamples and projected implementation on explicit and randomized finite designs.

7 Comparison with OCO-2

The companion paper [5] applies the same construction to the OCO-2 radiative-transfer emulation problem of Lamminpää et al. [4]: per spectral band (O_2 , weak CO_2 , strong CO_2), a reduced

atmospheric state of twenty or twenty-four coordinates maps to forty reduced radiance coefficients, with errors on the standardized coefficients and on the reconstructed radiance, under the ten-split protocol of Section 2. The corresponding results in [5] provide a comparison at a higher input dimension.

On this higher-dimensional problem, a kernel on the state trails the network by a factor of nine, four and five on the reduced coefficients: the isotropic Matérn reaches 40.6%, 58.6% and 43.4% on the three bands against 4.4%, 16.4% and 8.2% for the network. The residual correction adds no material gain in the reported OCO-2 comparisons. Little alignment with useful kernel directions is a possible explanation, but the public records do not measure the residual’s Gram-eigenvector coefficients. The design identity in Proposition 2 alone cannot establish that explanation on test data. The coordinate search that gives a kernel on the state its own length scales, a strong raw-input construction on the EMIT table, is not the strongest there either. Four learned metrics were run at the same ten splits on all three bands: a kernel-flow metric [24], a Mahalanobis metric under the same flow, the gradient outer product of the recursive feature machine [12], and per-input scales fitted by empirical Bayes. The last is the best of the four at every band, taking the isotropic kernel’s 40.6%, 58.6% and 44.3% on the coefficients to 18.7%, 25.3% and 16.9%, and on O₂ its radiance from 0.233% to 0.150%; the two flow metrics agree within half a point of each other at 26.9%, 29.8% and 23.9%, the recursive feature machine reads 28.2%, 47.9% and 31.3%, and an additive kernel 38.2%, 46.9% and 47.7%. The isotropic baseline agrees between the metric and baseline experiments on O₂ and weak CO₂ to two decimals; on strong CO₂ it reads 44.3% in the metric experiment and 43.4% in the baseline experiment. This band also has a spread of ± 3.4 because two of its ten splits land three points below the other eight. The learned metric reduces the isotropic error to between two fifths and a half of itself depending on the band. The feature kernel remains more accurate, with a band-dependent margin: the exact Matérn regression on the network’s frozen features reads 4.1%, 16.3% and 8.1% at the same splits, so the best state metric is 4.6 times that on O₂ but only 1.6 and 2.1 times it on the two CO₂ bands. The empirical-Bayes fit has substantial split variability: on O₂ the empirical-Bayes metric is bimodal across splits, eight of the ten landing near 17.2% and two at 24.7%, so its spread of ± 3.2 is thirty times the isotropic kernel’s ± 0.1 and its mean is not the value a single split is likely to show. The feature-kernel construction remains useful on both tasks: the exact Matérn regression on the network’s frozen features beats a ridge readout refitted on the same features at every split, on every band and under every training loss, and a per-coefficient choice among heads trained under different losses has lower mean error than the release’s stored emulator under both reported metrics. In the archived loss campaign it wins the reduced-coefficient comparison at all ten splits on each band and the radiance comparison at ten, nine and ten splits on O₂, weak CO₂ and strong CO₂, respectively. These companion comparisons are not an independent EMIT validation; their tables are in [5].

The OCO-2 retrieval records overlap the release’s test set. The reference’s data release carries 976 retrieval records, each with an atmospheric state, the full-physics monochromatic radiance at that state, its rank-40 reconstruction, and the prediction of the release’s Gaussian-process emulator at the same forty coefficients. Every one of the 976 reduced states matches a stored test input within 6.67×10^{-16} ; the records map to 974 distinct test rows, and the nearest test-row index is the same in all three bands. The records are therefore not disjoint from the stored test set and cannot serve as an independent evaluation pool, though they do provide a different target, the full-physics radiance, at those states. The state reduction was replayed independently from the release’s two dimensional-reduction files and the `x_in` and `geo_in` fields

of all 976 records; the replay code and its digests are archived with the per-run records.

8 Ten further emulation configurations

Table 9 extends the comparison to ten emulation configurations. Each family uses validation-based hyperparameter selection. The models and training budgets are specified below; these comparisons assess the selected family implementations rather than the best task-specific methods available for each corpus.

The corpora are the advection equation with discontinuous initial data from the operator suite of [16], which is also the source of the structural-mechanics problem of [5]; Burgers’ equation at three viscosities from PDEBench [17]; three one-step maps from The Well [18] — a two-dimensional turbulent radiative layer, an active-matter system, and a Helmholtz staircase; the ClimSim atmospheric physics emulation task [19]; and the paired correction-coefficient corpus of the physics-guided radiative-transfer work of [3], which is a third radiative-transfer emulation problem and the largest here at 487,790 rows. That last corpus is run in both of its configurations: from the atmospheric state alone, and with the low-fidelity 6S coefficients supplied as additional inputs, which is a different and easier problem.

The configurations are these. Advection is the 200-point input/output pair of the operator suite, used at full resolution with no reduction. Burgers is PDEBench’s `1D_Burgers_Sols_Nu` release, the map from the initial condition to the solution at $t = 1$ on a grid subsampled by four to 256 cells, the input used raw and the output reduced to 128 principal coefficients. The three Well datasets are one-step maps of the full multi-channel state with 256 principal coefficients in and out: the turbulent radiative layer on its 128×384 grid (density, pressure, velocity), active matter on 256×256 (concentration, velocity, and the two tensor fields), and the Helmholtz staircase on its two pressure channels. ClimSim is the low-resolution release, 124 inputs to 128 tendencies, with 100,000 training rows drawn from the 10.1M available and validation and test taken from the release’s own validation and scoring files. The correction-coefficient corpus is the paired 6S/libRadtran Sentinel-2 set, seven state variables and band wavelength mapping to the three libRadtran coefficients, split by atmospheric state so that all bands of a state travel together. The inputs are wavelength, solar zenith, view zenith, relative azimuth, aerosol optical depth, water vapor, ozone and elevation. The multi-fidelity configuration adds the three 6S coefficients, giving input dimensions eight and eleven, respectively. For these two configurations the error is

$$E_{0.05} = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i - \hat{Y}_i\|_2}{\max\{\|Y_i\|_2, 0.05\}}. \quad (15)$$

The same denominator floor enters validation selection and the kernel-flow objective. It limits the weight of small-norm coefficient vectors; these two rows therefore use a floored relative error rather than an unmodified relative norm.

The families are those of Section 3 with two differences forced by scale. The reference family is a ridge regression linear in whatever inputs the models receive, not the cubic of Table 1: a cubic expansion of a 256-dimensional input is not tractable, and the linear reference is used consistently for the additional configurations. And the kernel solves are exact only when the training block has at most 20,000 rows; above that they use a Nyström approximation on 6,000 uniformly drawn landmarks. That cap binds on ClimSim, on the Helmholtz staircase at 20,384 rows, and on the two correction-coefficient configurations. Those entries are what a capped fit gives; they are approximate rather than full dense solves. Their errors need not bound those of

corpus	n	d	q	runs	ridge (linear)	KRR	KRR, learned metric	network	network ensemble	network + residual KRR	kernel on features	convex stack
Advection	19,000	200	200	5	11.29±0.04	11.36±0.04	11.36±0.04	11.69±0.07	11.50±0.05	11.50±0.05	11.64±0.07	11.36±0.04
Burgers $\nu = 0.1$	7,200	256	128	3	13.89±0.53	2.81±0.28	2.90±0.33	20.34±0.61	19.28±0.09	2.78±0.29	4.92±0.30	2.78±0.29
Burgers $\nu = 0.01$	7,200	256	128	3	33.68±0.28	23.40±0.54	23.58±0.50	36.65±1.95	35.31±0.70	23.12±0.56	29.20±0.59	23.01±0.54
Burgers $\nu = 0.001$	7,200	256	128	3	38.38±0.12	31.66±0.16	33.23±1.73	39.98±0.65	38.89±0.11	31.82±0.11	35.59±0.48	31.34±0.14
Turbulent radiative layer	7,200	256	256	3	31.73	30.31	30.26±0.01	32.17±0.02	31.58±0.02	30.40±0.01	31.08±0.01	30.12
Active matter	14,000	256	256	1	37.24	35.81	35.18	37.31	36.93	35.77	36.22	35.05
Helmholtz staircase	20,384	256	256	1	0.00	0.26	0.07	0.40	0.24	0.23	0.45	0.07
ClimSim	100,000	124	128	5	32.24±0.06	27.20±0.05	26.31±0.04	25.10±0.18	24.43±0.17	24.33±0.13	24.71±0.25	24.25±0.17
RT corrections, state inputs	487,773–487,823	8	3	5	71.54±0.59	62.93±0.51	3.97±0.42	3.75±0.25	3.24±0.11	3.13±0.05	3.52±0.07	3.09±0.05
RT corrections, multi-fidelity inputs	487,773–487,823	11	3	5	33.83±0.26	15.76±1.98	4.60±0.31	2.79±0.14	2.40±0.11	2.34±0.09	2.80±0.09	2.32±0.08

Table 9: Test error [%] on ten further emulation configurations, mean $\pm$ standard deviation over the available runs. Errors are mean relative L^2 norms, except the two correction-coefficient rows, which use the denominator floor in (15). n is the number of examples — a range where the state-level split of the correction-coefficient corpus makes it differ by run — d the input dimension the models receive and q the output dimension they regress. The runs are random splits except for the turbulent radiative layer, whose three runs share The Well’s own split and differ in network initialisation, and the two single-run rows. The family set and validation-based selection rule are common across rows; the network budget is not, and is 200 epochs at width 512 except on ClimSim (60 at 512) and the two correction-coefficient configurations (120 at 384). The ridge column is linear in the inputs the models receive, unlike the cubic reference of Table 1, and is not a member of the convex stack; its Helmholtz-staircase entry is 0.004, below the two decimals shown. Kernel solves are exact when the training block has at most 20,000 rows and otherwise use 6,000 Nyström landmarks, which is the case on ClimSim, the Helmholtz staircase and the two correction-coefficient rows.

the full kernel, since approximation affects both fit and regularization. The convex stack and the per-coordinate selection are formed from the kernels, the networks, the residual correction and the feature kernel; the ridge is a reference row and is not one of their members.

What each row averages differs and is stated in the table. Advection, Burgers, ClimSim and the two correction-coefficient configurations have three to five random splits. The Well’s releases carry their own train, validation and test files, which are used unchanged, so the three turbulent-radiative-layer runs share one split and differ only in network initialisation; their spread, of order 0.01 points, measures initialisation and not sampling. The active-matter and Helmholtz-staircase rows are single runs.

The relative errors vary substantially with the problem. Where the map is smooth and the sample small — Burgers at $\nu = 0.1$ — the exact kernel is at 2.81% against the network’s 20.34%, a factor of seven, of the same kind as the five- to six-fold gap between the per-input kernel and the network on the EMIT table’s smooth components; as the viscosity falls and the solutions roughen the gap closes to 23.40% against 36.65% and then to 31.66% against 39.98%. Where the sample is large and the map rough — ClimSim, and the correction-coefficient corpus at nearly half a million rows — the network leads instead, 25.10% against 27.20% and 3.75% against 62.93%, which is the OCO-2 regime. Where the operator is linear, on advection, the linear ridge is the lowest row and every other family lands within 0.4 points of it, consistent with the linear form of the operator.

Learned metrics give particularly large improvements on the correction-coefficient corpus. On the correction-coefficient corpus from the state alone the isotropic kernel is at 62.93 ± 0.51 and a kernel-flow metric [24] takes it to 3.97 ± 0.42 , a factor of sixteen in the mean, which brings a kernel on the state to within six percent of the network on a corpus where the plain kernel was seventeen times worse. With the low-fidelity coefficients supplied as inputs the same pair reads 15.76 and 4.60, a factor of three: the easier the problem is made for the plain kernel, the less the learned metric adds. This comparison, like the EMIT and OCO-2 results, shows that

the choice of input metric can substantially change a kernel baseline.

The convex stack is at or near the best member of its candidate pool on each configuration. The convex stack is the lowest row on eight of the ten configurations and the second lowest on the other two; the residual-corrected network is in the lowest two on five. Both exceptions are the linear problems, and on both the row that beats the stack is the ridge, which the stack does not contain. On advection the ridge is lower by 0.07 points while the stack, at 11.36%, matches the best member it has — the isotropic kernel, on which it puts weight 0.87 — to two decimals. On the Helmholtz staircase the ridge reaches 0.004%, sixteen times below the next row of the table and about a hundred times below the weakest member of the pool, and the stack reaches 0.066%, again a shade below its own best member, the learned-metric kernel at 0.067%, on which it puts weight 0.94. In both cases the stack does what it is asked to do inside its pool; what it cannot do is recover a family the pool excludes. The statement this supports is correspondingly narrow: a combination is a good default when no member of its own pool dominates it, and a combination can only be read against the whole comparison when the pool contains the families the comparison includes.

The comparisons have several limitations. These rows use one training budget per corpus and one architecture family, and several of these corpora have published results from methods tuned for them specifically; the comparison here is between families under a common protocol, not against the state of the art on each problem. With three to five runs, and one on two rows, the spreads support the large orderings above and not fine distinctions between adjacent families. And every kernel column here inherits the tuning protocol Section 5.1 measures: the length scale is chosen on a 6,000-row subsample and the winner refit on every row, and this difference in sample size can affect the selected hyperparameters. On the EMIT table that costs the isotropic kernel a factor 0.882; on ClimSim, where the subsample is a sixteenth of the solve, re-choosing the scale against the solve that actually runs halves it and lowers the kernel’s error by 3.6%; on a 20,000-row corpus it changes nothing. These examples show that tuning can affect the comparison. They do not imply that every reported kernel error is pessimistic or that a reported gap is an upper bound: validation retuning can also increase test error. The columns of Table 9 describe the specified training and selection procedures, not the optimum attainable by either family.

9 Discussion

Exact kernel regression is effective on this six-dimensional radiative-transfer table. Input-specific length scales substantially reduce radiance error relative to the isotropic kernel, cubic regression and width-512 network. Residual fitting improves the network under the implemented PCA projection. The width-2000 network has larger standalone error under the tested training configuration, yet its features support more accurate kernel predictions at all ten splits. Since width and training duration change together, this comparison concerns the fitted representations rather than an intrinsic effect of width.

The central result is the difference between forward and inverse rankings. The input-scaled kernel has the smallest median reflectance error in the main comparison, whereas the feature kernels have smaller 95th-percentile inverse errors. Both narrow and wide convex stacks increase those tails at every split relative to their feature kernels. For the wide model, the mean radiance gain is only 0.0006 percentage points, while the mean within-split 95th-percentile reflectance error increases from 7.57 to 22.71 points. A ranking based only on mean radiance would obscure

this trade-off. The finite-perturbation identities show that inversion also requires control of the predicted denominator relative to component errors. Retrieval validation should specify a physical passband, report retained coverage and invalid inversions, and evaluate a range of surface reflectances.

The residual-error identity yields an exact alignment criterion for improvement in a quadratic metric. The PCA decomposition adds the truncation error explicitly. Applying a kernel learning-rate bound to the neural residual requires a uniform bound and a separate source-norm estimate; the observed learning slopes describe only the measured sample-size range. The OCO-2 and additional-corpus comparisons likewise support task-dependent choices of representation and kernel metric, rather than a single preferred model family.

Accuracy must also be considered alongside cost. Dense kernel regression stores order n^2 entries and requires order n^3 operations for factorization. One $18,884 \times 18,884$ double-precision matrix occupies approximately 2.85 GB before factorization workspaces and other arrays. Feature kernels evaluate both the network and the kernel against the training design; stacks evaluate each active member. Per-model throughput was not measured under matched conditions, so these results do not establish a speed advantage over operational emulators such as sRTMnet [2].

The ten partitions quantify within-table split sensitivity. Model development used the same table, and the partitions do not provide an untouched confirmation set. The two principal EMIT comparisons have complete per-split summaries; individual records are incomplete for some secondary ablations. Access to the raw arrays and simulator configuration is required to reproduce the training campaign. Within these limits, the results show that neural features and exact kernels can complement one another, while a small forward-error gain can coincide with a substantially worse inverse-error tail.

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Declaration of competing interest

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AI-assisted contributions

Claude Code (Anthropic) and Codex (OpenAI) implemented and ran the data pipeline, model comparisons, scaling experiments, diagnostics and figure generation, and drafted the manuscript

from the author’s data and the methodology of [5]. ChatGPT (OpenAI) developed the residual-error and inversion-stability arguments, checked the finite identities, analysed the stored per-split results, implemented the conditioning diagnostics and verification tests, and edited the manuscript.

Data and code availability

The manuscript, model drivers, run summaries and table-generation scripts are available at <https://github.com/yspennstate/emit-oco2-kernel-emulators>.

The width-512 and width-2000 EMIT experiments each contain ten per-split records with array digests, selected hyperparameters, split sizes and evaluation metrics. These twenty records reproduce Tables 1, 7, 2 and 3. Reproduction commands and the file inventory are given in the repository documentation.

Secondary learning-curve, width and tuning summaries contain ten-run aggregates. The public individual archive lacks one isotropic and one input-scaled record at 8,000 rows; it contains three width-1000 records and five full-data-tuning records. The three-member wide-feature comparison uses three splits, and width 4,000 uses two. The corresponding aggregate tables and individual records are distinguished in the reproduction manifest.

The EMIT arrays, identified as `data_EMIT_24k` and attributed to the Jet Propulsion Laboratory, are not redistributed. Raw arrays, trained weights and prediction files are available from the author on request. The public archive does not specify the simulator version, generation configuration or redistribution permissions. Scoring uses float64; prediction archives stored as float32 cannot reproduce cancellation-sensitive inverse errors at the same precision. The conditioning utility requires explicit approval of precision-altered inputs. Its tests use synthetic arrays; the EMIT results reported here are all-band statistics rather than conditioned evaluations.

The OCO-2 arrays and reference predictions are from the release accompanying [4]. The additional configurations use the cited public releases.

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